1)

$0.5 \times \tan(0.4 \times 1.75) = 0.421$ units^2 (3 SF)

2) a)

$h = 2$

	0		2
$x_{i-\frac{1}{2}}$		1	
e^{-x^2}		e^{-1^2}	

$\int_0^2 e^{-x^2}\,dx \approx 2\left(e^{-1^2}\right) = 0.735759$ (6 DP)

2) b)

$h = 1$

	0		1		2
$x_{i-\frac{1}{2}}$		0.5		1.5	
e^{-x^2}		e^{-1^2}		$e^{-1.5^2}$	

$\int_0^2 e^{-x^2}\,dx \approx 1\left(e^{-0.5^2} + e^{-1.5^2}\right) = 0.884200$ (6 DP)

2) c)

$h = 0.5$

	0		0.5		1		1.5		2
$x_{i-\frac{1}{2}}$		0.25		0.75		1.25		1.75	
e^{-x^2}		$e^{-0.25^2}$		$e^{-0.75^2}$		$e^{-1.25^2}$		$e^{-1.75^2}$	

$\int_0^2 e^{-x^2}\,dx \approx 0.5\left(e^{-0.25^2} + e^{-0.75^2} + e^{-1.25^2} + e^{-1.75^2}\right) = 0.882789$ (6 DP)

3) a)

	1		2
$x_{i-\frac{1}{2}}$		1.5	
$\frac{1}{1-\ln x}$		$\frac{1}{1-\ln 1.5}$	

$\int_1^2 \frac{1}{1-\ln x}\,dx \approx 1\left(\frac{1}{1-\ln 1.5}\right) = 1.681987$ (6 DP)

3) b)

	1		1.5		2
$x_{i-\frac{1}{2}}$		1.25		1.75	
$\frac{1}{1-\ln x}$		$\frac{1}{1-\ln 1.25}$		$\frac{1}{1-\ln 1.75}$	

$\int_1^2 \frac{1}{1-\ln x}\,\mathrm{d}x \approx 0.5\left(\frac{1}{1-\ln 1.25}+\frac{1}{1-\ln 1.75}\right) = 1.778992$ (6 DP)

3) c)

	1		1.25		1.5		1.75		2
$x_{i-\frac{1}{2}}$		1.125		1.375		1.625		1.875	
$\frac{1}{1-\ln x}$		$\frac{1}{1-\ln 1.125}$		$\frac{1}{1-\ln 1.375}$		$\frac{1}{1-\ln 1.625}$		$\frac{1}{1-\ln 1.875}$	

$\int_1^2 \frac{1}{1-\ln x}\,\mathrm{d}x \approx 0.25\left(\frac{1}{1-\ln 1.125}+\frac{1}{1-\ln 1.375}+\frac{1}{1-\ln 1.625}+\frac{1}{1-\ln 1.875}\right) = 1.809250$ (6 DP)

4) a)

	0		0.5		1		1.5		2
$x_{i-\frac{1}{2}}$		0.25		0.75		1.25		1.75	
$\frac{\sin x}{x}$		$\frac{\sin 0.25}{0.25}$		$\frac{\sin 0.75}{0.75}$		$\frac{\sin 1.25}{1.25}$		$\frac{\sin 1.75}{1.75}$	

$\int_0^2 \frac{\sin x}{x}\,\mathrm{d}x \approx 0.5\left(\frac{\sin 0.25}{0.25}+\frac{\sin 0.75}{0.75}+\frac{\sin 1.25}{1.25}+\frac{\sin 1.75}{1.75}\right) = 1.61$ (3 DP)

4) b)

Increase the number of strips used.

5)

	0		2		4		6		8		10
$t_{i-\frac{1}{2}}$		1		3		5		7		9	
v		12.8		17.2		20		21.2		20.8	

$\int_0^{10} 10+3t-0.2t^2\,\mathrm{d}t \approx 2(12.8+17.2+20+21.2+20.8) = 184$ metres

6) a)

	1		2		3		4
$x_{i-\frac{1}{2}}$		1.5		2.5		3.5	
$\ln x$		$\ln 1.5$		$\ln 2.5$		$\ln 3.5$	

$\int_1^4 \ln x \ \mathrm{d}x \approx 1(\ln 1.5 + \ln 2.5 + \ln 3.5) = 2.5745$ (4 DP)

6) b)

$u = \ln x, \frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{x}, \frac{\mathrm{d}v}{\mathrm{d}x} = 1, v = x$

$\int_1^4 \ln x \ \mathrm{d}x = [x \ln x]_1^4 - \int_1^4 \frac{x}{x} \ \mathrm{d}x = [x \ln x - x]_1^4 = (4 \ln 4 - 4) - (\ln 1 - 1) = -3 + 4 \ln 4$

6) c)

$\frac{2.5745-(-3+4\ln 4)}{-3+4\ln 4} = 0.0115$

7) a)

$\int_a^b \mathrm{f}(x)\,\mathrm{d}x \approx (b-a)\mathrm{f}\left(\frac{a+b}{2}\right)$

7) b)

The tangent line PQ lies on the curve at the midpoint $x = \frac{a+b}{2}$

The line has equation has gradient $\mathrm{f}'\left(\frac{a+b}{2}\right)$

Let $\mathrm{f}'\left(\frac{a+b}{2}\right) = m$

So that in general:

$y - \mathrm{f}\left(\frac{a+b}{2}\right) = m\left(x - \left(\frac{a+b}{2}\right)\right)$

At $x = a$ then $y - \mathrm{f}\left(\frac{a+b}{2}\right) = m\left(a - \left(\frac{a+b}{2}\right)\right) = m\left(\frac{a-b}{2}\right)$ so $y = m\left(\frac{a-b}{2}\right) + \mathrm{f}\left(\frac{a+b}{2}\right)$

At $x = b$ then $y - \mathrm{f}\left(\frac{a+b}{2}\right) = m\left(b - \left(\frac{a+b}{2}\right)\right) = m\left(\frac{b-a}{2}\right)$ so $y = m\left(\frac{b-a}{2}\right) + \mathrm{f}\left(\frac{a+b}{2}\right)$

Area under the tangent line:

$\frac{1}{2} \times \left(m\left(\frac{a-b}{2}\right) + \mathrm{f}\left(\frac{a+b}{2}\right) + m\left(\frac{b-a}{2}\right) + \mathrm{f}\left(\frac{a+b}{2}\right)\right)(b-a)$

$= \frac{1}{2} \times \left(m\left(\frac{a-b}{2}\right) + \mathrm{f}\left(\frac{a+b}{2}\right) - m\left(\frac{a-b}{2}\right) + \mathrm{f}\left(\frac{a+b}{2}\right)\right)(b-a)$

$= \mathrm{f}\left(\frac{a+b}{2}\right)(b-a)$

This is the same as the mid-ordinate rule with one strip.

Since this trapezium is an under-estimate, the mid-ordinate rule estimate must be an under-estimate.

Alternatively:

The tangent line PQ touches the curve at the midpoint $x = \frac{a+b}{2}$

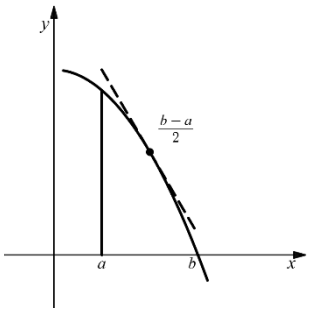
The height of the trapezium at the midpoint is $\mathrm{f}\left(\frac{a+b}{2}\right)$

The tangent is linear and symmetric about the midpoint, so the mean of the endpoint heights equals the midpoint height

$\therefore \mathrm{f}\left(\frac{a+b}{2}\right)(b-a)$ which is the same as the mid-ordinate rule with one strip.

Since this trapezium is an under-estimate, the mid-ordinate rule estimate must be an under-estimate.

8)



$\int_a^b \mathrm{f}(x)\,\mathrm{d}x \approx (b-a)\mathrm{f}\left(\frac{a+b}{2}\right)$

Suppose there is a tangent line which touches the curve at the midpoint $x = \frac{a+b}{2}$

The height of the trapezium at the midpoint is $\mathrm{f}\left(\frac{a+b}{2}\right)$

The tangent is linear and symmetric about the midpoint, so the mean of the endpoint heights equals the midpoint height

$\therefore \mathrm{f}\left(\frac{a+b}{2}\right)(b-a)$ which is the same as the mid-ordinate rule with one strip.

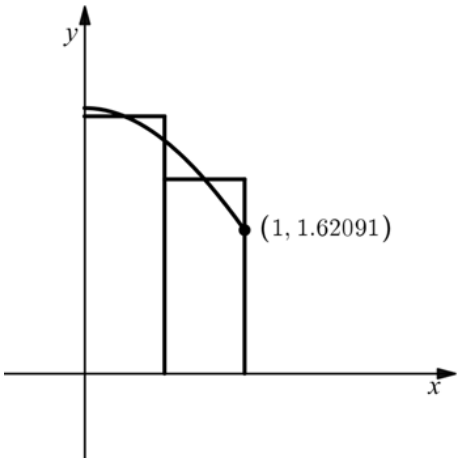
Since this trapezium is an over-estimate, the mid-ordinate rule estimate must be an over-estimate.

9) a)

	0		0.5		1
$x_{i-\frac{1}{2}}$		0.25		0.75	
$3\cos x$		$3\cos 0.25$		$3\cos 0.75$	

$\int_0^1 3\cos x\ \mathrm{d}x \approx 0.5(3\cos 0.25 + 3\cos 0.75) = 2.55$ (3 SF)

9) b)



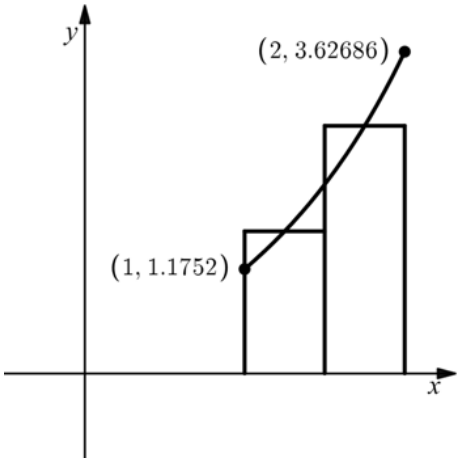
The curve is concave (concave downwards) for $0 \leq x \leq 1$ so the estimation is an over-estimate.

10) a)

	1		1.25		1.5		1.75		2
$x_{i-\frac{1}{2}}$		1.125		1.375		1.625		1.875	
$\sinh x$		$\sinh 1.125$		$\sinh 1.375$		$\sinh 1.625$		$\sinh 1.875$	

$\int_1^2 \sinh x \ \mathrm{d}x \approx 0.25(\sinh 1.125 + \sinh 1.375 + \sinh 1.625 + \sinh 1.875) = 2.21$ (2 DP)

10) b)



The curve is convex (concave upwards) for $1 \leq x \leq 2$ so the estimation is an under-estimate.

11) a)

	1	2	3	4
$\sqrt{x^2-1}$	0	$\sqrt{3}$	$\sqrt{8}$	$\sqrt{15}$

$\int_1^4 \sqrt{x^2-1}\,\mathrm{d}x \approx \frac{1}{2}\times 1\left(0+\sqrt{15}+2\left(\sqrt{3}+\sqrt{8}\right)\right) = 6.4970$ (4 DP)

11) b)

	1		2		3		4
$x_{i-\frac{1}{2}}$		1.5		2.5		3.5	
$\sqrt{x^2-1}$		$\frac{\sqrt{5}}{2}$		$\frac{\sqrt{21}}{2}$		$\frac{3\sqrt{5}}{2}$	

$\int_1^4 \sqrt{x^2-1}\,\mathrm{d}x \approx 1\left(\frac{\sqrt{5}}{2}+\frac{\sqrt{21}}{2}+\frac{3\sqrt{5}}{2}\right) = 6.7634$ (4 DP)

11) c)

Since the curve is concave (concave downwards), the trapezium rule under-estimates the area.

11) d)

Since the curve is concave (concave downwards), the mid-ordinate rule over-estimates the area.

12) a)

	2	4	6	8	10
$\frac{1}{\ln x}$	$\frac{1}{\ln 2}$	$\frac{1}{\ln 4}$	$\frac{1}{\ln 6}$	$\frac{1}{\ln 8}$	$\frac{1}{\ln 10}$

$\int_2^{10} \frac{1}{\ln x}\,\mathrm{d}x \approx \frac{1}{2}\times 2\left(\frac{1}{\ln 2}+\frac{1}{\ln 10}+2\left(\frac{1}{\ln 4}+\frac{1}{\ln 6}+\frac{1}{\ln 8}\right)\right) = 5.3977$ (4 DP)

12) b)

	2		4		6		8		10
$x_{i-\frac{1}{2}}$		3		5		7		9	
$\frac{1}{\ln x}$		$\frac{1}{\ln 3}$		$\frac{1}{\ln 5}$		$\frac{1}{\ln 7}$		$\frac{1}{\ln 9}$	

$\int_2^{10} \frac{1}{\ln x}\,\mathrm{d}x \approx 2\left(\frac{1}{\ln 3}+\frac{1}{\ln 5}+\frac{1}{\ln 7}+\frac{1}{\ln 9}\right) = 5.0012$ (4 DP)

12) c)

$\mathrm{f}'(x) = \frac{0-\frac{1}{x}}{(\ln x)^2} = \frac{-1}{x(\ln x)^2}$

$\mathrm{f}''(x) = \frac{(\ln x)^2+2(\ln x)}{x^2(\ln x)^4} = \frac{\ln x+2}{x^2(\ln x)^3}$

Since $\ln x + 2 > 0$ for all $x > 0$ and $x^2(\ln x)^3 > 0$ for $x > 1$ and so in the interval $2 \leq x \leq 10$ we have that $\mathrm{f}''(x) > 0$

So $\mathrm{f}'(x)$ is always increasing for $2 \leq x \leq 10$

12) d)

Since the curve is convex (concave upwards), the trapezium rule over-estimates the area.

12) e)

Since the curve is convex (concave upwards), the mid-ordinate rule under-estimates the area.

13) a)

	1	2	3
$\frac{\mathrm{e}^x}{x}$	e^1	$\frac{\mathrm{e}^2}{2}$	$\frac{\mathrm{e}^3}{3}$

$\int_1^3 \frac{\mathrm{e}^x}{x}\,\mathrm{d}x \approx \frac{1}{2}\times 1\left(\mathrm{e}^1+\frac{\mathrm{e}^3}{3}+2\left(\frac{\mathrm{e}^2}{2}\right)\right) = 8.40126$ (5 DP)

13) b)

	1		2		3
$x_{i-\frac{1}{2}}$		1.5		2.5	
$\frac{e^x}{x}$		$\frac{e^{1.5}}{1.5}$		$\frac{e^{2.5}}{2.5}$	

$\int_1^3 \frac{e^x}{x}\,dx \approx 1\left(\frac{e^{1.5}}{1.5}+\frac{e^{2.5}}{2.5}\right) = 7.86079$ (5 DP)

13) c)

The trapezium rule under-estimates the area for $1 \leq x \leq 3$ and the mid-ordinate rule over-estimates the area for $1 \leq x \leq 3$, so the curve is convex (concave upwards) for $1 \leq x \leq 3$

14) a)

$\pi \int_0^2 (\ln(x+1))^2\,dx$

14) b)

	0		0.5		1		1.5		2
$x_{i-\frac{1}{2}}$		0.25		0.75		1.25		1.75	
$(\ln(x+1))^2$		$(\ln(1.25))^2$		$(\ln(1.75))^2$		$(\ln(2.25))^2$		$(\ln(2.75))^2$	

$\pi \int_0^2 (\ln(x+1))^2\,dx \approx 0.5\pi((\ln(1.25))^2+(\ln(1.75))^2+(\ln(2.25))^2+(\ln(2.75))^2)$
$= 3.21$ units^3 (3 SF)

15) a)

	0		0.5		1
$x_{i-\frac{1}{2}}$		0.25		0.75	
$\sin(x^2)$		$\sin(0.25^2)$		$\sin(0.75^2)$	

$\int_0^2 \sin(x^2)\ \mathrm{d}x \approx 0.5(\sin(0.25^2) + \sin(0.75^2)) = 0.2979$ (4 DP)

15) b)

$\mathrm{f}(x) = 3\sin((x-2)^2)$

15) c)

$\int_2^3 \sin(x^2)\ \mathrm{d}x \approx 3 \times 0.2979 = 0.894$ (3 DP)

15) d)

	2		2.5		3
$x_{i-\frac{1}{2}}$		2.25		2.75	
$3\sin((x-2)^2)$		$3\sin(0.0625)$		$3\sin(0.5625)$	

$\int_0^2 \sin(x^2)\ \mathrm{d}x \approx 0.5(3\sin(0.0625) + 3\sin(0.5625)) = 0.894$ (3 DP)

16) a)

	0		1		2		3
$x_{i-\frac{1}{2}}$		0.5		1.5		2.5	
$\sqrt{81-3x^3}$		$\frac{\sqrt{1290}}{4}$		$\frac{9\sqrt{14}}{4}$		$\frac{\sqrt{546}}{4}$	

$\int_0^3 \sqrt{81-3x^3}\ \mathrm{d}x \approx 1\left(\frac{\sqrt{1290}}{4} + \frac{9\sqrt{14}}{4} + \frac{\sqrt{546}}{4}\right) = 23.24$ (2 DP)

16) b)

$f(x) = (81 - 3x^3)^{\frac{1}{2}}$ so $f'(x) = -\frac{9}{2}x^2(81 - 3x^3)^{-\frac{1}{2}}$

The tangent line:

$f'(2) = -\frac{6\sqrt{57}}{19}$

$y - \sqrt{57} = -\frac{6\sqrt{57}}{19}(x - 2)$

$19y - 19\sqrt{57} = -6\sqrt{57}x + 12\sqrt{57}$

$19y + 6\sqrt{57}x = 31\sqrt{57}$

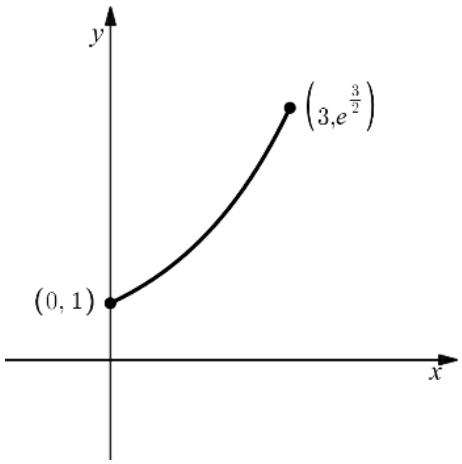
16) c)

When $x = 0$, $y = \frac{31\sqrt{57}}{19}$

When $y = 0$, $x = \frac{31}{6}$

So the area is $\frac{1}{2} \times \frac{31\sqrt{57}}{19} \times \frac{31}{6} - 23.24 = 8.58$ units^2 (3 SF)

17) a)



17) b)

	0		1		2		3
$x_{i-\frac{1}{2}}$		0.5		1.5		2.5	
$e^{\frac{x}{2}}$		$e^{\frac{1}{4}}$		$e^{\frac{3}{4}}$		$e^{\frac{5}{4}}$	

$\int_0^3 e^{\frac{x}{2}}\,dx \approx 1\left(e^{\frac{1}{4}} + e^{\frac{3}{4}} + e^{\frac{5}{4}}\right) = 6.891368$ units^2(6 DP)

17) c)

$\frac{1}{b-a}\int_a^b \mathrm{f}(x)\,\mathrm{d}x = \frac{1}{3}\int_0^3 \mathrm{e}^{\frac{x}{2}}\,\mathrm{d}x \approx \frac{1}{3}\times 6.891368 = 2.29712$ (5 DP)

18) a)

$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3}$

18) b)

	0.1		0.2		0.3		0.4		0.5
$x_{i-\frac{1}{2}}$		0.15		0.25		0.35		0.45	
$x - \frac{x^2}{2} + \frac{x^3}{3}$		0.139875		0.223958		0.303042		0.379125	

$\int_{0.1}^{0.5} x - \frac{x^2}{2} + \frac{x^3}{3}\mathrm{d}x \approx 0.1(0.139875 + 0.223958 + 0.303042 + 0.379125) = \frac{523}{5000}$

19) a)

$x\mathrm{e}^x = x\left(1 + x + \frac{x^2}{2}\right) = x + x^2 + \frac{x^3}{2}$

19) b)

	0		0.05		0.1		0.15
$x_{i-\frac{1}{2}}$		0.025		0.075		0.125	
$x + x^2 + \frac{x^3}{2}$		0.0256328		0.0808359		0.141602	

	0.15		0.2		0.25		0.3
$x_{i-\frac{1}{2}}$		0.175		0.225		0.275	
$x + x^2 + \frac{x^3}{2}$		0.208305		0.28132		0.361023	

$\int_0^{0.3} x + x^2 + \frac{x^3}{2}\mathrm{d}x$

$\approx 0.05(0.0256328 + 0.0808359 + 0.141602 + 0.208305 + 0.28132 + 0.361023)$

$= 0.0549$ (4 DP)